



Security architecture for automotive communication networks with CAN FD

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ABSTRACT

Modern vehicles have dozens of electronic modules, connected through sophisticated intra-vehicle networks, which essentially control all the vehicle functions. Recent studies have shown the vulnerability of vehicular computer systems, but the existing security models are not enough robust yet because the busload and level of cryptographic strategy are not considered enough to be implemented in the frame. This work proposes a new security model for the Controller Area Network with Flexible Data-Rate (CAN FD), based on encryption and authentication. The model divides the frame data field into two blocks. The AES-256 (Advanced Encryption Standard) symmetric key encryption is applied to the first block of the data frame, whose content is the data corresponding to the payload of the frame. The authentication of the data in the first block is executed through the implementation of the HMAC/SHA-256 function and inserted in the second block. Performance analysis has shown that the combination of 32 bytes for the first block and 32 bytes for the second one provides the shortest processing time and 33.33% fewer data compared to frame 48. Results indicated that the proposed security model has a higher encryption performance than existing methods.

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1. Introduction

Over the past twenty years, embedded technologies have incorporated several new functionalities in vehicles, to improve, for example, fuel consumption, safety, and comfort [Navet et al. \(2005\)](#). As a result, the current vehicles carry intricate and complex networks with dozens of electronic control units (ECUs). Those systems contain millions of code lines that run into tens of heterogeneous processors, connected by internal networks, such as the Controller Area Network (CAN) [Backhausen \(2017\)](#). Modern-day vehicles can no longer be perceived as just mechanical systems, with over 100 million lines of code in the overall architecture, higher than a modern operating system or a Boeing 757 [El-Rewini et al. \(2020\)](#). Another important serial data communication protocol for automotive safety-critical applications is the FlexRay. It enables real-time time communication, high bandwidth, event and time trigger communication, fault tolerance, and provides deterministic

communication for automotive applications [Bao \(2009\)](#) [Vaz et al. \(2020\)](#). However, the lack of intrinsic security mechanisms makes FlexRay vulnerable to spoofing and DoS attacks while existing features are not sufficient to assure reliable communication in the presence of adversaries [Murvy and Groza \(2020\)](#). Even a complex communication technology can have also vulnerabilities that allows cyber attacks [Murvy and Groza \(2019\)](#).

The field of automotive electronics is characterized by a large number of interconnected electronic devices with a wide range of features, manufactured by several suppliers. The current systems in automotive and industrial fields contain complex functions that demand CAN-based higher-layer protocols with high data throughput. CAN technology provides bandwidth of up to 1 Mbps and a payload of up to 8 Bytes, which is currently overloaded and cannot address the demand for the next generation of automotive electronic systems. Despite several proposed alternatives, much effort has been focused on the enhancement of CAN, which has been upgraded to Controller Area Network with Flexible Data Rate (CAN-FD). Therefore, the CAN-FD is an improved communication technology that provides high bandwidth of up to 8 Mbps and a payload of up to 64 Bytes [de Andrade et al. \(2018\)](#). There have been

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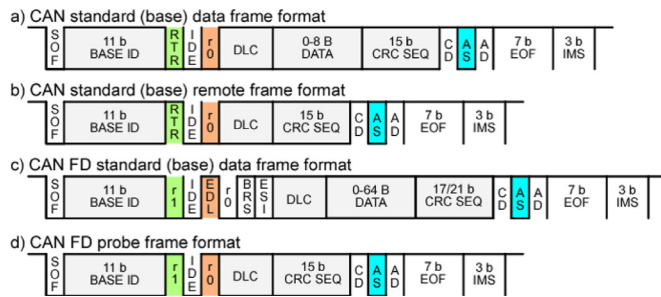


Fig. 1. CAN and CAN FD Frames format.

also some researches about a new protocol, which can mix the bus with CAN FD, called CAN XL, which uses Automotive Ethernet technology and reaches a rate of 20 Mbps and a payload of 2048 bytes. It should also be mentioned that the next generation of vehicles will likely have only a few powerful ECUs Bosch (2022). The recent development of the Control Area Network standard is also presented based on CAN with Flexible Data rate (CAN FD). V2X communication security is analyzed based on four categories of attacks: 1) authenticity/identification; 2) availability; 3) data integrity; and 4) confidentiality. Moreover, the blockchain based security measures for vehicular network are related on the literature Gao et al. (2022). In communication systems, faults affect the transmission medium and related circuits Checkoway et al. (2011), as they can corrupt transmitted messages. To recover from those situations, some networks have implemented several fault-tolerant mechanisms. However, this creates a communication overhead by introducing delivery delays in messages, leading to performance degradation in the control system Bernardini et al. (2017). When messages have real-time requirements, which is common in control systems, these issues can disturb severely the system operation and can even lead to its failure Zhang et al. (2014).

In addition to in-vehicle communication, new communication types have been developed over the last few years, such as between two vehicles (labeled as V2V) or between a vehicle with road infrastructure (V2I). The implementation of these innovations can lead to more efficient and safer transportation systems, paving the way for autonomous vehicles Houmansadr et al. (2011); Ghorri et al. (2018). However, it is important to observe that hackers can exploit vulnerabilities in any of those connected systems. In particular, modern vehicles have become targets, vulnerable to malicious attacks. Previous works have shown that an attacker connected to the internal network of a car can circumvent all computer control systems, including safety-critical components, such as brakes and engines Checkoway et al. (2011); Bernardini et al. (2017); Zhang et al. (2014); Miller and Valasek (2013); He et al. (2018). In 2016, Volkswagen (VW) vehicles were discovered to have flaws with their Remote Keyless Entry (RKE) systems that would enable attackers to unlock VW vehicles. That same year, the members of the Keen Security Lab, a division of the Chinese firm Tencent, demonstrated their ability to remotely hijack the brakes of a Tesla Model S vehicle. Two years later, the same lab released a report identifying vulnerabilities within BMW vehicles that would allow attackers to access the CAN bus and send illegitimate CAN messages El-Rewini et al. (2020). There are several ways that hackers can access the vehicle, for example, by Bluetooth wireless connections, OBD II physical connections, or even from a virus in a smartphone connected to the vehicle Houmansadr et al. (2011). Fig. 1 shows that the frame of CAN FD (in a simplified version) almost does not have any security system. The frame is divided into CAN ID which is the identifier with 11 bits in standard mode, the control field that prepares the data before being transmitted, the payload with 512 bits, the Cyclic Redundancy Check (CRC) that calculates if there is an invalid byte transmitted with 17 or 21 bits,

and the acknowledge (ACK) that validates if the bit was correctly delivered, with 2 bits.

While some research about the vulnerability of vehicle systems has been reported in the literature, there has been little awareness of the extent of potential risks, which are still considered excessively unrealistic for concern Onishi (2012); Wang and Liu (2018). Recently, some works have proposed security strategies to mitigate attacks on vehicles. For example, some authors Woo et al. (2016) proposed a new frame CAN FD model as a countermeasure based on another recent research Miller and Valasek (2013). Another work proposed a new security architecture for the communication between ECUs on different channels through gateway ECU (GECU) Agrawal et al. (2019). The results also showed that using an authenticated encryption scheme has better performance than applying individual primitives for encryption and authentication Agrawal et al. (2019). A survey classified current cryptographic and IDS approaches and compared them based on criteria associated with real-time constraints, types of hardware used, changes in CAN Bus behavior, types of attack mitigation, and software/hardware used to validate these approaches Carel et al. (2018) Aliwa et al. (2021). An IDS is proposed based on a Deep Convolutional Neural Network (DCNN) to protect the controller area network (CAN) bus of the vehicle. The DCNN learns the network traffic patterns and detects malicious traffic without hand-designed features. The experimental results demonstrate that the proposed IDS has significantly low false negative rates and error rates when compared to the conventional machine-learning algorithms, increasing also the complexity of the system Kosmanos et al. (2020). Other authors designed a protocol based on efficient symmetric primitives, taking advantage of two procedures: splitting keys between nodes and mixing authentication tags Groza et al. (2017). Machine Learning (ML), Neural Networks (NN) and Deep Learning (DL) algorithms have been widely used to detect malicious traffic and classify network attacks. Various examples that use Deep NN can be found in the literature. Another scheme utilizing anomaly detection using a Graph Neural Network (GNN) is a technique in study, that interconnections of the involved nodes were taken into account in the computed adjacency matrix, along with various topological characteristics of the graph Hidalgo et al. (2022). Another paper proposed a novel graph-based Gaussian naive Bayes (GGBN) intrusion detection algorithm by leveraging graph properties and PageRank-related features. The GGBN on the real rawCAN data set yields 99.61%, 99.83%, 96.79%, and 96.20% detection accuracy for denial of service (DoS), fuzzy, spoofing, replay, mixed attacks, respectively Islam et al. (2022) and another one proposed a CAN firewall, namely CANSentry, that prevents malicious nodes' misbehaviors such as injecting unauthorized commands or disabling targeted services Humayed et al. (2020), while another one approaches about a delay based in plug-in-monitor for IDS in CAN Wang et al. (2018a).

This work presents a model to improve the protection of security systems for vehicles with CAN-FD. The proposed solution uses an existing HMAC (Keyed-Hash Message Authentication Code)/SHA-2 (Secure Hash Algorithm 2) for data authentication and AES-256 for data encryption. The authentication and cryptography techniques are inserted in the data frame, aiming to reach the best frame in terms of security, processing time, and with a lower bus-load compared to other works. Also, data from a CAN Bus of a real vehicle was collected to use the real data as the input of the security schemes.

The Controller Area Network with Flexible Data Rate (CAN FD) is an enhance on the CAN 2.0 as defined in the ISO 11898-1 standard, where it presents new features that enable higher speeds than traditional CAN when transmitting the data part, but is designed to be backward compatible. The physical and data link layers can be shared by the message formats of both protocols in such

a way that any CAN FD node can receive messages sent from a CAN node. This property allows for a smooth transition from CAN to CAN FD. We consider the specifications in, which define a format different from ISO CAN FD. Therefore, CAN FD frame allows the transmission of payloads up to 64 bytes, as opposed to the maximum 8 bytes of CAN and an increased baud rate of up to 8 Mbit/s for the bits following the arbitration stage (the message identifier).

This work aims to build a frame model for CAN FD considering its intrinsic vulnerability. Although some research on the vulnerability of vehicle systems has been reported in the literature, there has been little awareness of the extent of potential risks [Onishi \(2012\)](#) [Wang and Liu \(2018\)](#).

The main contributions of this work are:

- We present some vulnerabilities of the communication protocols of vehicular networks employing the statistical analysis of messages from the database extracted from a CAN Bus vehicle network;
- We develop a frame model based on existing strategies and encryptions that can mitigate these attacks; and
- We propose a new CAN FD frame that provides data authentication and a good performance to encrypt and decrypt data.

This work is structured as follows: Section II describes the theoretical background and Section III presents the security mode. Section IV presents the results, and Section V discusses the conclusion and potential future works.

2. Theoretical background

Malicious attacks could be of several forms since vehicle systems could be accessed by physical connections or wireless in the ECUs. This section addresses the main threat models to a modern vehicle and the algorithms currently used for authentication and cryptography. The ECU structure can be subdivided into the input signal conditioning, the logical processing of those signals, the microcomputer and logic levels of output power, and a control signal or command [Backhausen \(2017\)](#); [Wang and Liu \(2018\)](#).

2.1. Security attacks in vehicles

Previous works have identified several possible attack scenarios, where various components have been altered or manipulated by external interventions [Miller and Valasek \(2013\)](#). Attacks to the odometer, speedometer, indicator lights horn operated, windshield wipers, door lock, electric glass drives, or even the shutdown of the engine and the driver brakes are the most known ones [Traub et al. \(2017\)](#). Therefore, considering the major attacks cataloged previously [Lukasiewicz et al. \(2016\)](#) [Xie et al. \(2021\)](#), attacks are made to manipulate the firmware of the ECU authenticated by the auto manufacturer. Moreover, the attacker is limited to polynomial computational power and storage; therefore, current cryptography including advanced encryption standards (AES) and keyed-hash message authentication code (HMAC) are secure against the attack.

There are seven types of access on CAN FD bus [Bernardini et al. \(2017\)](#). The first one is the physical access that directly or indirectly provides access to their internal networks. Access is granted through intermediary physical interfaces. The second one is the OBD II, commonly accessed during routine maintenance by technicians, mechanics, and service providers to perform diagnostics and code the ECUs [Wang et al. \(2018b\)](#) [Mohammad et al. \(2016\)](#). The third one is entertainment, which supported several audio formats (such as MP3 and WMA). Similarly, vehicle manufacturers also provide some sort of external digital multimedia port (usually a USB port or a connection to other devices) to allow users to control the media of the car. As a result, a hacker can attack through the CD

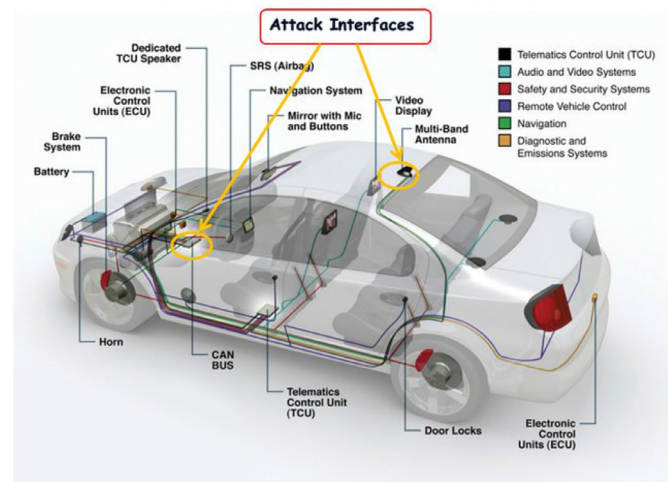


Fig. 2. The attack interfaces of CAN bus in a vehicle [Murvay and Groza \(2019\)](#).

input with malicious codes [Cho and Shin \(2016\)](#). Thus, a compromised CD player can provide an effective way to attack other automotive components [Bernardini et al. \(2017\)](#). Fourth, wireless access allows malicious connections that can be established via a wireless transmitter in the car receiver's proximity to a range between 5 and 300 meters (short-range) and can reach distances greater than 1 km, using digital access channels (long-range). Fifth, Bluetooth with a range of around 10 meters, it can be extended through amplifiers and directional antennas [Bernardini et al. \(2017\)](#). Sixth, the RFID, such as the vehicle key that incorporates an RFID tag on a key ring or key and a reader near the car's steering column [Mohammad et al. \(2016\)](#). Seventh, the streaming channels, which are not specifically intended for a given automobile but can be "tuned" by receptors on demand.

External attack connections, such as long-range transmission, can be attractive as control channels (i.e. for triggering attacks) since they can carry out commands to both receptors and do not require accurate information addressing their targets [Bernardini et al. \(2017\)](#). Modern cars have a multitude of transmission channels, such as the Global Positioning System (GPS), digital radio, and Radio Data System (RDS). It is possible to observe in [Fig. 2](#), extracted from [Murvay and Groza \(2019\)](#), all discussions in the paragraphs before about connected ECU's and their vulnerabilities.

2.2. SHA-256 HMAC Algorithm and AES-256

The HMAC (Hash-based Message Authentication) is an algorithm for generating a cryptographically secure MAC from a secret key for any message. The Hash (key + message) algorithm is highly recommended, in order to have more protection for information. The HMAC can be expressed as:

$$HMAC(K, m) = H(K \oplus opad \parallel H(K \oplus ipad \parallel m)) \quad (1)$$

where $\oplus$ denotes the bitwise xor, it means when two inputs are the value of the same bit, the output is 0, otherwise, it is 1. Now, $\parallel$ can connect two pieces of data, K is the key, m is the input message, $opad$ and $ipad$ are parameters of external and internal hash (for SHA-256 $opad$ and $ipad$ are 0x36 and 0x5C, respectively, repeated 64 times results) [Onishi \(2012\)](#).

After 2010, the National Institute of Standards and Technology (NIST) recommended using SHA-2 instead of SHA-1. Therefore, here, we used SHA-2, more specifically SHA-256, which can convert input data to create output with 256 bits of length. The Advanced Encryption Standard (AES), also known by its original name Rijndael AES, is a variant of the Rijndael block cipher with three

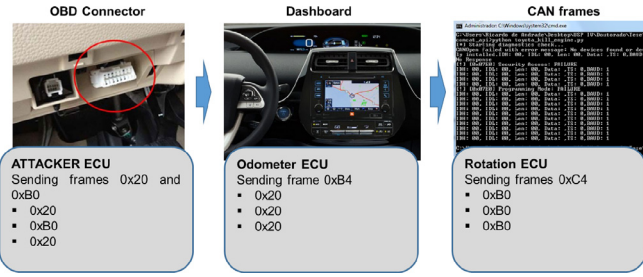


Fig. 3. Inserting Fake Data as a CAN Bus Attack.

different key lengths: 128, 192, and 256 bits [Rao et al. \(2017\)](#); [Daemen and Rijmen \(2013\)](#).

3. Security model for can fd

In this section we show an, since a study of case, hacking a CAN bus to a criteria to choose a new security frame of the CAN FD Bus.

3.1. Specific can bus attack to show vulnerabilities on automotive network

Figure 3 shows two legitimate ECUs and one ECU attacker that creates and injects messages with forged ID, DLC, and data. This attack aims to override any periodic message sent by a legitimate safety-critical ECU so that their receiver ECUs get distracted or become inoperable. In this example, the attacker ECU 0xN ("N" here means that it is possible to be any ID and spoofing the bus) injects several attack messages with values equal to 20 and B0, usually sent by a legitimate ECU 0xB4 (18ms) and 0xC4 (28ms) at a frequency of 1MHz. Then, other nodes that typically receive message B0 are forced to receive the attack messages more often than the legitimate ones. Demonstrated attacks such as controlling vehicle maneuvers and monopolizing the CAN bus with the highest priority messages exemplify a fabrication attack.

3.2. Setting of can fd bus - case study

This work used a hybrid vehicle to create a database of a CAN Bus in order to convert it into a CAN FD bus observed in Fig. 5 and an attacked architecture with CANoe and VN1630 in Fig. 6 that could be used as the data for the performance analysis in Section V. We use the development kit (ECOM Cable) from EControls, which allowed a sniff (data scanning) of the CAN network, shown in Fig. 4.

This vehicle has two CAN networks, both operating at a data rate of 500 kbps. The network connection can be made via OBD II (pins 6 and 14) or by directly connecting the diagnostic tool to

CAN High and CAN Low. The Engine Control Module (ECM) receives all the nodes of other modules via the CAN network, such as the transmission module, airbag, anti-skid control, acceleration, rotation, and energy [de Andrade et al. \(2018\)](#). In any CAN bus system, there are some modules interconnected with each other through the OBD. Due to the connection with the OBD, it ends up making it easy for unauthorized people to access the bus. This allows attackers to make some crucial modifications to the vehicle, such as turning off the engine, turning off ABS brakes, powering the Air Bag, as well as other actions, but there are some modules that can not be sniffed using only OBD II access point. In order to do that, it is necessary to insert exploits directly on the CAN Bus.

After laboratory acquisitions using the VN 1630 Diagnostic Tool with Vector CANoe 16 software, it was possible to map the data that travels in the vehicle. To make the physical connection of the cables, it was necessary to find out which ECU wire is CAN High 1 and CAN Low 1, and CAN High 2 and CAN Low 2. It shows the possibilities that hackers have when accessing a vehicle with a bus CAN. Applying the strategy proposed here, data from the database in Table are converted to CAN FD using Eq. 2 [de Andrade et al. \(2018\)](#). Having obtained the WCTT (Worst-case Transmission Time) of each message τ_m , and given the period of each message ω_m , the system busload can be obtained by summing up the busload of each message:

$$\beta = \sum m \frac{\tau_m}{\omega_m} \quad (2)$$

and $\tau = T_s + T_f$, where T_s are the slow bits on the frame and T_f are the fastest bits [de Andrade et al. \(2018\)](#). The amount of bytes that can be inserted in the data field of the CAN FD frame is defined by the DLC (Data Length Control) field, and the allowed data sizes are as follows: 8, 12, 16, 20, 24, 32, 48, and 64 bytes. Then, it is considered the maximum size of the CAN FD data frame of 64 bytes and divide it into two blocks of lengths h_1 and h_2 , and implement encryption and authentication, according to Eq. 3:

$$h_1 + h_2 = 64 \quad (3)$$

where h_1 is the length of the encrypted data block and h_2 is the length of the authentication block.

3.3. Security model architecture

Both CAN and CAN-FD do not have security functions, such as authentication, encrypted data, or access control. In authentication, a receiver identifies a data stream in the data frame only with the transmitter ID, which causes the data messages to be sent back to the bus with information manipulation. The CAN and CAN-FD have the CRC field that only identifies an error in data transmission, but fails when the message is sent again. These two networks, as they

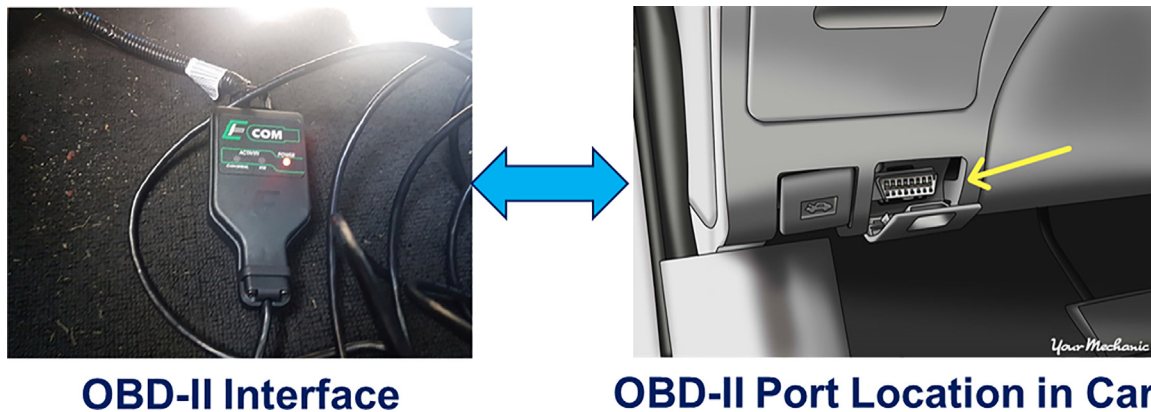


Fig. 4. ECOM cable connected on OBD-II.



Fig. 5. A hybrid vehicle being attacked, and used to create a database.

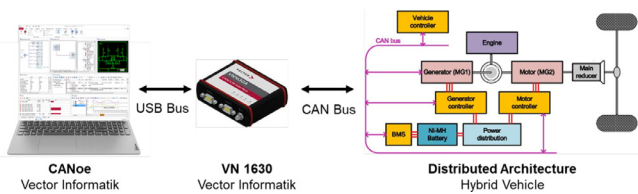


Fig. 6. Attacked architecture with CANoe and VN1630.

do not have encrypted data and send messages to the entire bus, end up being easily "spied on".

This work proposes a novel CAN FD frame with encryption and data authentication, as shown in Fig. 8. The new fields, named Cipher Data with h_1 bytes and Auth Data with h_2 bytes, are inserted into the payload of the frame. There are some options to implement the new frame according to CAN FD configuration Manivannan et al. (2020), as shown in Table 2. It is possible to observe that Data Field with 64 bytes can be divided into seven possible frames. It is important to notice that $h_1 + h_2 = 64$. This work aims to find the best combination of h_1 and h_2 , in order to develop a security model, as shown in Fig. 8.

An architecture schematic is illustrated in Fig. 7. The blocks in green color show how cryptography and authentication can work in a vehicle with CAN FD. This security model architecture uses three keys as one of the main building blocks for secure embedded communication: session, temporary, and factory keys. The factory key is a static one (like a MAC address) assigned by the manufacturer and it is a security component for each ECU to do full protection to automotive distributed computer system. Each ECU has 1 (one) factory key. The session key is used for encryption and authentication and must be updated in periodic intervals.

For secure communication, a CAN FD security protocol should offer a secure and fast key distribution mechanism for data frame encryption and authentication. Moreover, an efficient and secure session key update protocol is needed to enhance the security of the session key. It is also needed to support the connectivity between an external device and a vehicle. For secure session key updates, three properties should be ensured based on [19,28]:

1) Forward and backward secrecy: In CAN FD, every ECU should use an authentication session key and an encryption session key to ensure the authentication and confidentiality of CAN FD data frames in the n th session.

2) Key freshness: The encryption and authentication keys, which are used for encryption and authentication of CAN FD data

frames in the n th session, should be freshly generated. To ensure key freshness, a parameter used for key generation should be continually changed using a random number or counter.

3) Temporary key: The temporary key is used as input for the generation of new session keys. It is necessary to use fresh key material that is re-distributed periodically. If only static keys were used, attacks against these (never changing) keys would break the system for the rest of its lifetime. A new session key is generated using as input the temporary key XOR factory key. Therefore, the proposed security architecture is also secured against replay attacks.

At the first moment, when the driver turns the car on, a temporary key is generated using a random number. An XOR operation with the factory key is performed to generate a session key for the cluster (instrumental panel in orange block) or ECU. This session key encrypts the data using AES 256-CBC in the green block called Cipher AES 256. After that, in another green block, an authenticator, called Authenticator (SHA-256) HMAC, calculates the integrity of the data, forming a CAN FD frame. Then, the data is sent to the transceiver and consequently to the CAN FD bus. When a module (ECU) receives the frame, the data is deciphered using the same algorithm AES 256, and then authenticated, using the same session key. In case of a session key update, a new random number is generated for the temporary key, then another XOR operation with the factory key. The interval can be set up every few minutes, for example.

From Algorithm 1, it is possible to select the best frame among the seven possibilities from Table 2, considering the busload, pay-

Algorithm 1 CRITERIA TO CHOOSE BEST FRAME.

Require: Payload(Frame48,Frame32,Frame24, Frame20,Frame16,Frame12,Frame8),Busload, time[En+De+SHA256(En/De)*2]

```

1: for each payload do
2:   if time ≤ 5ms then
3:     if busload ≤ 98, 86% then
4:       if payload ≥ 24 then
5:         Select : Payload
6:       end if
7:     end if
8:   end if
9: end for
10: Output = Best Frame;
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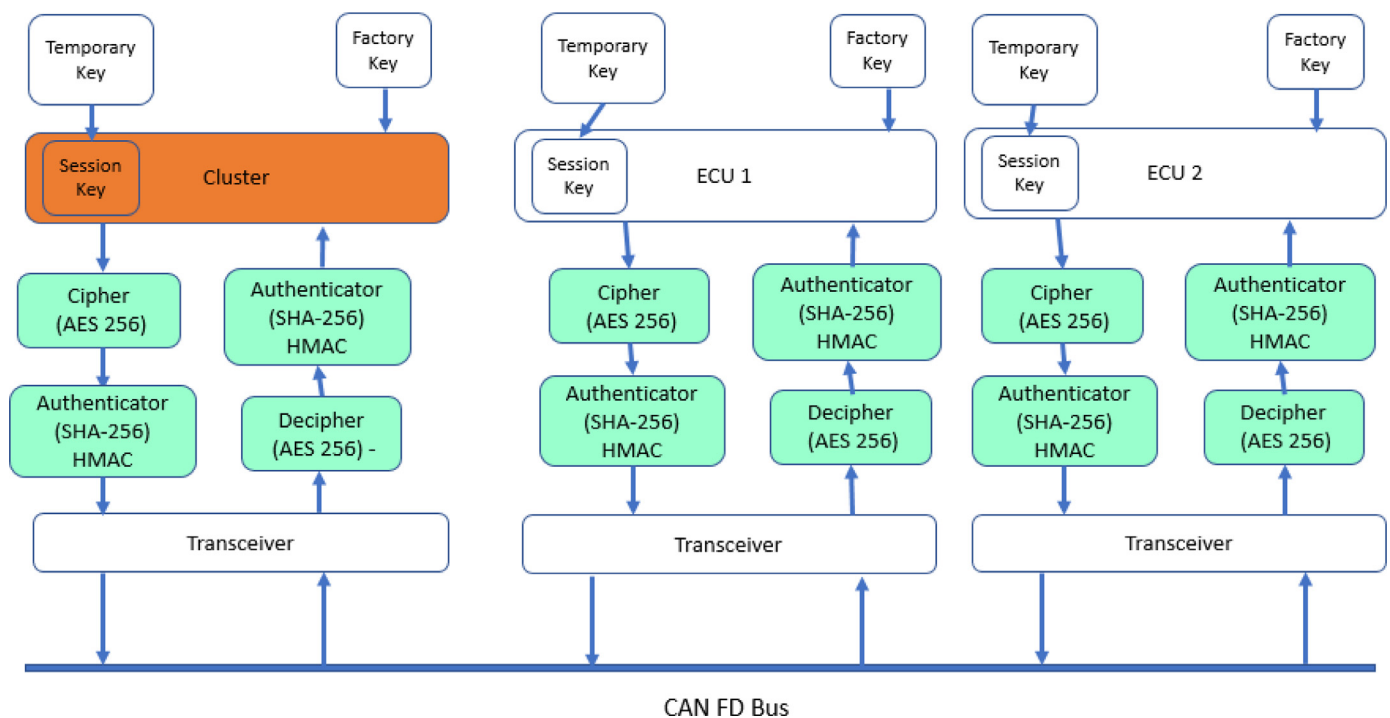


Fig. 7. Architecture of the proposed distributed cryptography model.

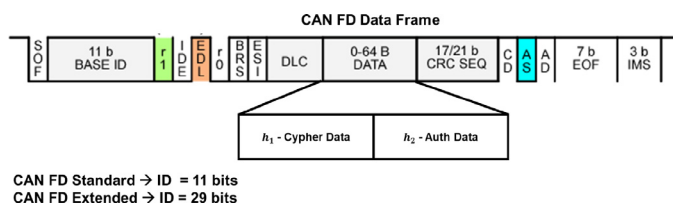


Fig. 8. Architecture of Payload's hypothesis.

load size, and time to cipher, decipher and authenticate. It is necessary to multiply the HMAC time by 2 since there is the time to generate the HMAC in the transmitter and the time to verify the authentication in the receiver. Furthermore, a time shorter than 5 ms and a busload smaller than 98,86% are important, since numbers above those values may generate overhead in the bus, according to [de Andrade et al. \(2018\)](#); [Woo et al. \(2016\)](#). Data payload must be at a minimum of 24 bytes since that with large demands on the bus, vehicles need at least this number of data bytes.

4. Development and results

All seven cases explored in this investigation are shown in Fig. 11 and Fig. 12, which illustrates the number of encrypted bytes, network load, and delays in sending messages. In these graphs, the analysis of the time curves refers only to the encryption and decryption time of frames 48, 32, 24, 20, 16, and 8 bytes.

4.1. Embedded software

We used an Arduino Mega 2560 working at 16 MHz, according to Figs. 9 and 10, in order to develop the test cases for the seven possible CAN FD frames. The tests consist of encryption and decryption of messages (from Table 1) using the AES256 library from Arduino with lengths of 48, 32, 24, 20, 16, 12, and 8 bytes (from Table 2) using AES 256 and SHA 256. Each measurement was performed one hundred times and the results in this section are the

average values. It is possible to observe in Figs. 11 and 12 the result of time to SHA 256 when there are 32 bytes of input data.

4.2. Results

All seven cases explored in this investigation are presented in Fig. 11, which shows the number of encrypted bytes, network load, and delays in sending messages. According to Fig. 11, the best strategy choice is when encrypted and decrypted data have the shortest time with the maximum number of bytes transmitted and at the minimum busload possible. Also, Fig. 11 shows the total time for encryption and decryption, considering a common base of 1000 bytes for data transmission. The encryption and decryption time is the least in frame 48 because few frames are being transmitted. In the case of 48, there are 20 complete frames, while in

Table 1
Frames of a Hybrid Vehicle.

ID	EVENT TYPE	DLC	DATA LENGHT	DATA
20	CAN Frame	7	7	00 00 07 01 00 00 2F
230	CAN Frame	7	7	00 02 00 05 00 00 40
25	CAN Frame	8	8	00 01 00 00 78 78 78 96
24	CAN Frame	8	8	02 01 02 06 61 FE 80 16
32A	CAN Frame	2	2	00 2F
AA	CAN FRAME	8	8	1A 6F 1A 6F 1A 6F 1A 6F
320	CAN Frame	8	8	20 20 00 00 F2 08 C0 25

Table 2
STRATEGY FOR DATA ENCRYPTED + HMAC.

DATA ENCRYPTED	HMAC
48	16
32	32
24	40
20	44
16	48
12	52
8	56

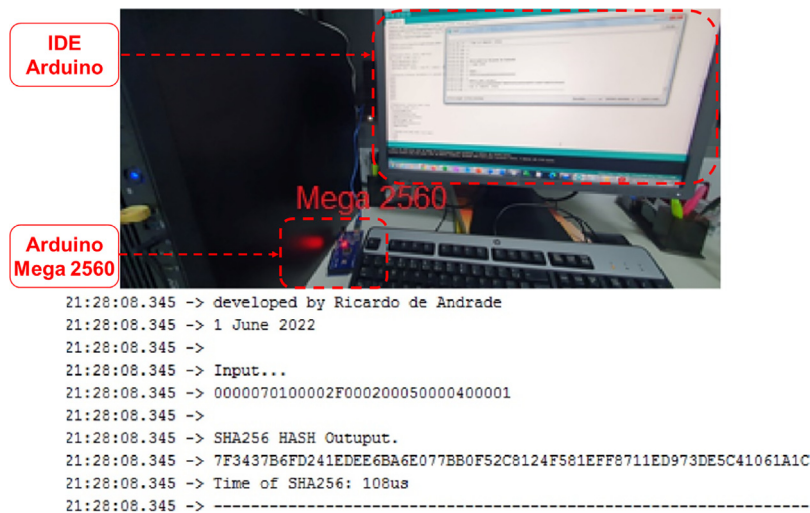


Fig. 9. Measurement of time to SHA256 -32 bytes.

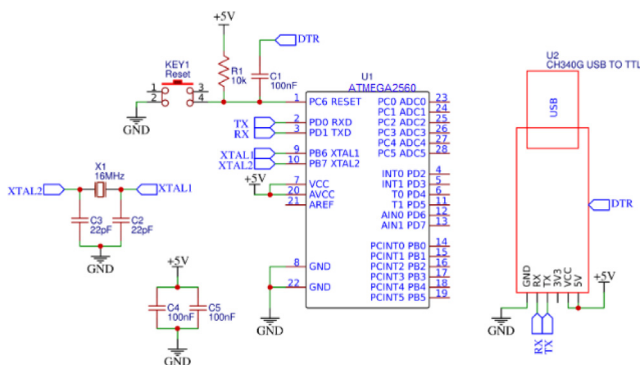


Fig. 10. Arduino's Electric Scheme.

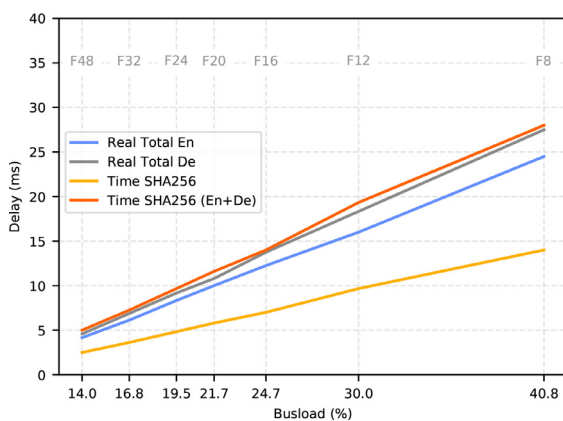


Fig. 11. Results of 7 possibilities of Frame and the best choice using a base of 1000 bytes.

the 32 frames there are 31 fragments. Thus, it is possible to conclude that the best frame case for CAN FD is in the $h_1 = 32$ and $h_2 = 32$ configuration since in this step the delay was 0.532 ms (AES En+ AES De +SHA 256) by frame, about one less than the model 48-16. Besides, if only the question of time is taken into account, frame 48 can encrypt and decrypt faster than the others, approximately 12.22% faster. It also takes into account the time to decipher to be the same. However, what is suggested in this research is to implement frame 32, which represents 33.33% fewer

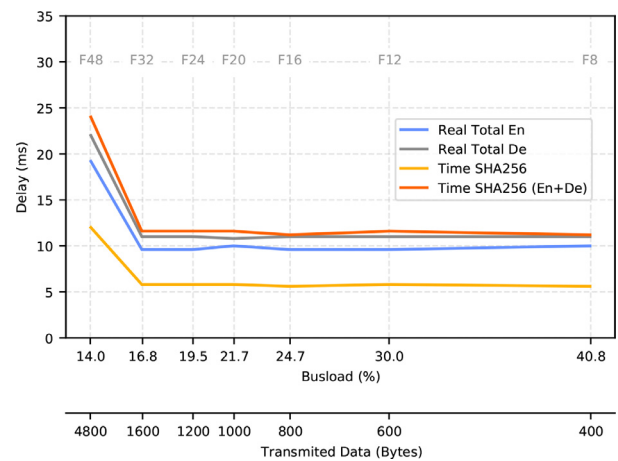


Fig. 12. Results of 7 possibilities of Frame and the best choice using a base common of 1 second.

data per frame on the bus. The time varies depending on the size of the data field, the busload becomes larger the greater the number of messages. It is important to notice the frames are showed as F48,F32,F24,F20,F16,F12,F8 on the grid's line of graphics.

Fig. 12 shows another metric, which is the relationship between the number of bytes sent per second. It is possible to observe that frame 48 sends 4800 bytes in 1 second, while frame 32 sends 1600 bytes, frame 24 sends 1200 bytes, and so on. When considering the time to cipher data by frame, the best case is frame 32, however, in the case of (Real total En +De frame), frame 48 is better than the others. Now, in case of decipher data, both frames (48 and 32) have the same time, while frame 48 is faster in the total decipher case. Then, frame 32 has a busload lesser than frame 48 in Fig. 12 and almost the same in Fig. 11. This means that sending a great quantity of data with a low busload and time is possible. It is important to notice that both Fig. 11 and Fig. 12 have [Real En by frame, Real De by frame, and SHA 256 (En/De)] in blank on the graph because the values are almost zero.

5. Conclusions and future work

In summary, this work has explored the security of the CAN FD bus and presented a security control for data encryption and

key exchange to protect data transmission. This research is important for the global automotive industry, as well as for other industries, which use either CAN or CAN FD bus, as it improves the communication protocol, preventing attackers from exploring vulnerabilities in the protocol. This work has proposed a new frame model that can protect the integrity and confidentiality of the data against external attacks. Moreover, the time to encrypt and decrypt data will be around 0.196 ms and 0.220 ms to a frame of 32 bytes.

As potential future work, a new integrated system using the proposed frame could be developed. Also, a CAN bus with integration into TCP/IP network systems is an important issue in the near future. Since society demands more connected devices, car-makers need to invest more in security to offer greater protection and data privacy in vehicles, particularly in the case of smart cities.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Ricardo de Andrade: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Supervision, Validation, Writing – review & editing. **Max Mauro Dias Santos:** Visualization, Investigation. **João Francisco Justo:** Data curation, Writing – original draft. **Leopoldo Rideki Yoshioka:** Data curation, Writing – original draft. **Hans-Joachim Hof:** Visualization, Investigation. **João Henrique Kleinschmidt:** Conceptualization, Methodology, Software.

Data availability

No data was used for the research described in the article.

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